**Table 3.10 Electronic configuration of elements in a period**

| Period number (n) | Filling of electrons in orbitals | | Number of elements | Outer shell Electronic configuration | |
|---|---|---|---|---|---|
| | Starts from | Ends with | | First element | Last element |
| 1 | 1s | 1s | 2 | H – $1s^1$ | He – $1s^2$ |
| 2 | 2s | 2p | 8 | Li – $2s^1$ | Ne– $2s^2 2p^6$ |
| 3 | 3s | 3p | 8 | Na – $3s^1$ | Ar – $3s^2 3p^6$ |
| 4 | 4s | 4p | 18 | K – $4s^1$ | Kr– $4s^2 4p^6$ |
| 5 | 5s | 5p | 18 | Rb – $5s^1$ | Xe – $5s^2 5p^6$ |
| 6 | 6s | 6p | 32 | Cs – $6s^1$ | Rn – $6s^2 6p^6$ |
| 7 | 7s | 7p | 32 | Fr – $7s^1$ | Og – $7s^2 7p^6$ |

In the fourth period the filling of 3d orbitals starts with scandium and ends with zinc. These 10 elements are called first transition series. Similarly 4d, 5d and 6d orbitals are filled in successive periods and the corresponding series of elements are called second, third and fourth transition series respectively.

In the sixth period the filling of valence electrons starts with 6s orbital followed by 4f, 5d and 6p orbitals. The filling up of 4f orbitals begins with Cerium (Z=58) and ends at Lutetium (Z=71). These 14 elements constitute the first inner-transition series called Lanthanides. Similarly, in the seventh period 5f orbitals are filled, and it's -14 elements constitute the second inner-transition series called Actinides. These two series are placed separately at the bottom of the modern periodic table.

### 3.4.2 Variation of Electronic Configuration in the Groups:

Elements of a group have similar electronic configuration in the outer shell. The general outer electronic configurations for the 18 groups are listed in the Table 3.11. The groups can be combined as s, p, d and f block elements on the basis of the orbital in which the last valence electron enters.

The elements of group 1 and group 2 are called s-block elements, since the last valence electron enters the ns orbital. The group 1 elements are called alkali metals while the group 2 elements are called alkaline earth metals. These are soft metals and possess low melting and boiling points with low ionisation enthalpies. They are highly reactive and form ionic compounds. They are highly electropositive in nature and most of the elements imparts colour to the flame. We will study the properties of these group elements in detail in subsequent chapters.



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